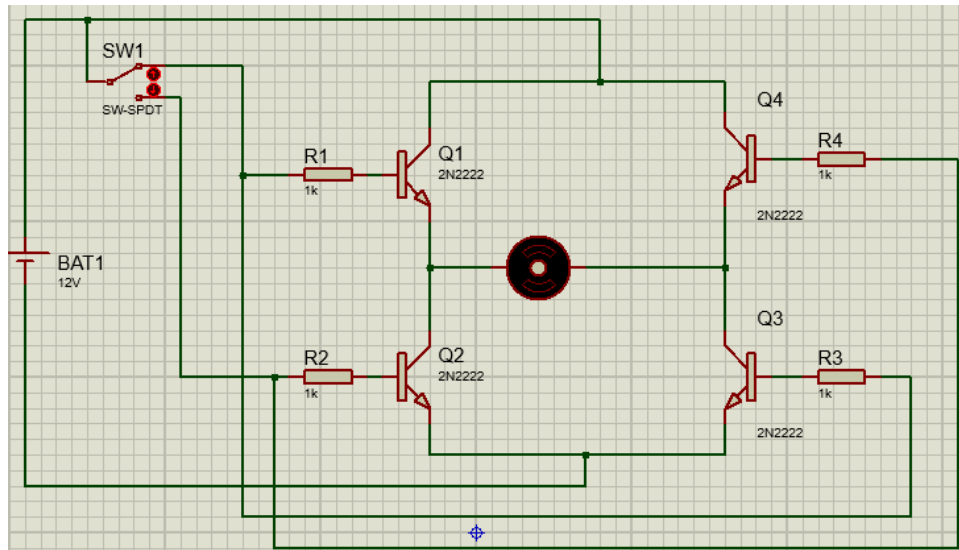


Electronics Session Tasks 1 And 2

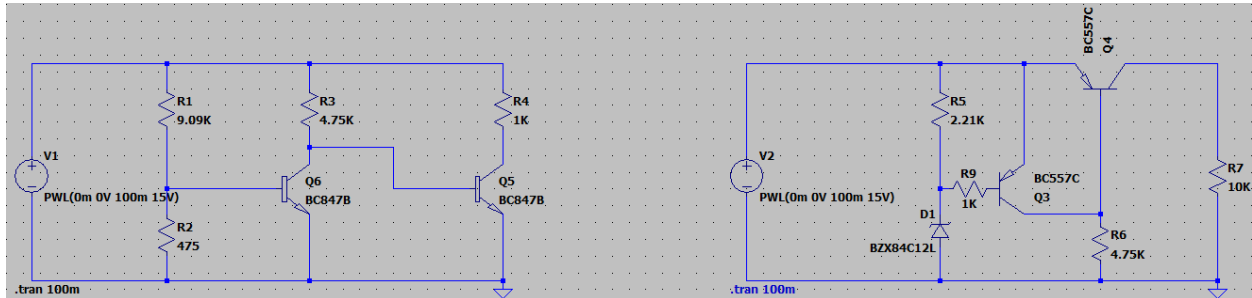
- Task 1



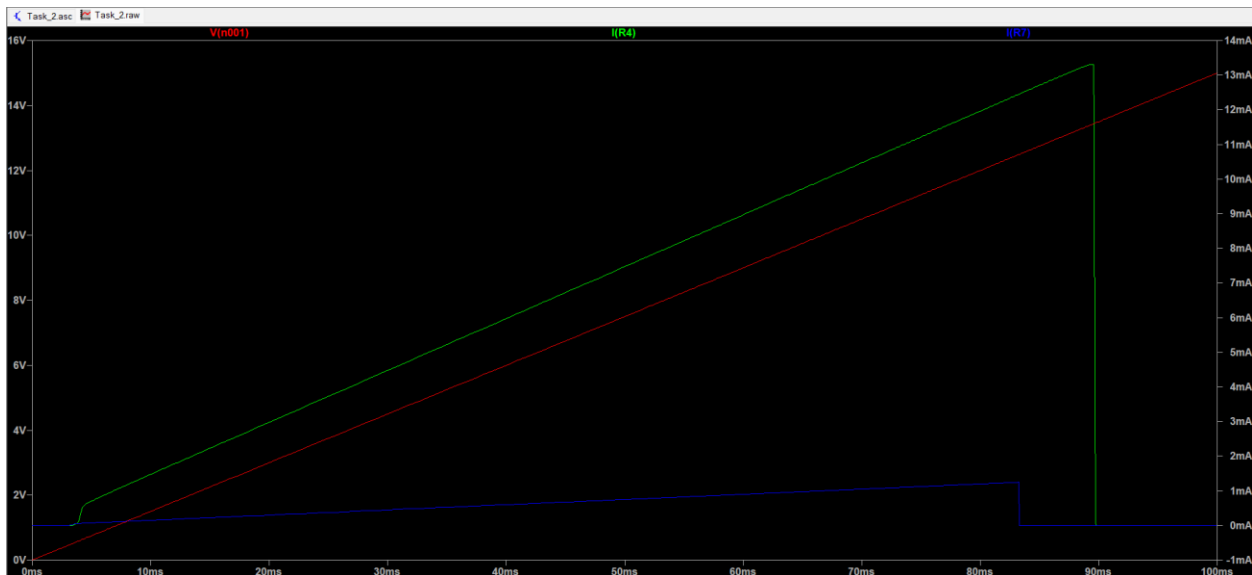
The above picture shows an H-bridge. The idea behind it is to keep it efficient yet as simple as possible. For this reason, four NPN transistors were used to control the current flow and, thus, control the motor's direction: the upper left (Q1) and lower right (Q3) are responsible for the motor's anticlockwise direction while the lower left (Q2) and the upper right (Q4) are responsible for the motor's clockwise direction. For each two transistors responsible for a motion direction, the other two transistors must be in the off state. To ensure that, a switch (SW1) is put directing electricity flow from the 12V battery, making this possible.

• Task 2

Circuit Analysis



Picture 1 shows an NPN BJT (on the left) and a PNP BJT (on the right).



Picture 2 shows input voltage in both circuits (red), the output current of the NPN left circuit on R4 (Green), and the output current of the PNP right circuit on R7 (Blue).

❖ Left: NPN stage (Q6→Q5, BC847B) — low-side Darlington current sink

What it is

- Two NPNs cascaded (Darlington-like). Q6 provides gain and drives Q5; Q5 pulls current **down** through the load **R4 (1 kΩ)** to ground.
- R1–R3–R2 form the base-bias network for Q6; Q6 then drives Q5's base.

How it behaves (green trace I(R4))

- As the input (red) ramps, the base of Q6 rises. When $V_{BE1} + V_{BE2} \approx 1.2\text{--}1.4\text{ V}$ is reached, the pair turns on.
- Collector current through **R4** is then roughly

$$I_{R4} \approx (V_{IN} - V_{CE}(Q5)) / R4$$

so, it increases almost linearly with V_{IN} (the green straight line).

- The sharp drop near ~90 ms is where the bias network stops supplying enough base drive (the small-signal Q6 saturates/steals drive), so the pair snaps off and the R4 current collapses.

Takeaways

- Very high current gain, nice linear rise with V_{IN} , but needs ~1.2–1.4 V headroom (two V_{BE}) and can “collapse” once the bias chain can't sustain base current.
- ❖ Right: PNP stage (Q3→Q4, BC557C → BCP57C) — high-side pass with Zener reference

What it is

- A PNP driver (Q3) feeding a larger PNP pass device (Q4). The Zener **D1 (≈12 V)** fixes the **base** potential; R9 biases the Zener; R5–R6 provide bias/feedback.
- Load is **R7 (10 kΩ)** to ground; current measured is I(R7) (blue).

How it behaves (blue trace I(R7))

- For low V_{IN} , Q3/Q4 are on, sourcing current to R7, so I_{R7} is >0.
- As V_{IN} climbs and the Zener conducts, the base of the PNPs is held near V_Z . The emitter rises with V_{IN} , so V_{EB} shrinks.

- When $V_{IN} \approx V_Z + V_{BE}$ ($\approx 12.6\text{--}13.0\text{ V}$), V_{EB} falls below conduction and the PNP pair turns **off**. That's why the blue current ramps slightly then drops to ~ 0 around $\sim 83\text{--}85\text{ ms}$.

Cutoff threshold = $V_Z + 0.6 - 0.7\text{ V}$

Takeaways

- Acts like a simple linear **high-side regulator/limiter**: it will source current until V_{IN} approaches $V_Z + V_{BE}$, then it cleanly shuts off.

Results Analysis

- **Why the slopes look different**
- **Green (NPN)**: roughly a straight line with slope $\approx 1\text{ mA/V}$ (set by $1\text{ k}\Omega$), because the transistor is pulling the node low current is basically V_{IN} / R_4 until base drive runs out.
- **Blue (PNP)**: small current because **R_7 is $10\text{ k}\Omega$** , and it stops entirely once the Zener-set threshold is reached. It's not trying to be linear with V_{IN} ; it's acting like an **OVP/threshold regulator**.
- **Headroom/compliance**

- **NPN** needs $\sim 2 \times V_{BE}$ to start, then can sink current over the whole input range (good compliance) until it hits a drive limit.
- **PNP** needs $V_Z + V_{BE}$ of headroom; **beyond that it purposely turns off.**

- **Use-case summary**

- **Left (NPN):** a high-gain **current sink / low-side switch**—good for converting a small control voltage into a larger load current (linear with V_{IN}).
- **Right (PNP):** a **high-side pass + Zener reference**—good for **clamping or cutting off** the load above a set input threshold.